Volume I. First Section. Rational Functions.

§ 9. Interpolation.

We shall make an application of the last obtained formulae to a problem from the theory of interpolation.

A whole rational function of degree n is to be determined which, for $n+1$ given values of the variable x, has prescribed values.

Thus the problem is the determination of the $n+1$ coefficients $a_0, a_1, \dots, a_n$ in

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n \tag{1}$$

from the $n+1$ equations

$$f(\alpha_0) = A_0, \quad f(\alpha_1) = A_1, \quad \dots, \quad f(\alpha_n) = A_n, \tag{2}$$

when $\alpha_0, \alpha_1, \dots, \alpha_n$ are the given values of x, and $A_0, A_1, \dots, A_n$ are the corresponding values of $f(x)$.

Equations (2) form a system of equations of the first degree for the unknowns $a_0, a_1, \dots, a_n$, which in general can be solved by determinants, as we shall see in the next section. We shall later return to this problem, but treat it now under the special supposition that the given values of x are the first $n+1$ whole numbers $0, 1, 2, \dots, n$.

The binomial coefficients $B_\nu^{(x)}$, as they were defined by § 7, (6), keep their good sense even when x is not a whole number, as was previously presupposed, but an arbitrary variable quantity. Then

$$B_\nu^{(x)} = \frac{x(x-1)\cdots(x-\nu+1)}{1\cdot 2\cdot 3\cdots \nu} \tag{3}$$

is a whole rational function of degree ν in x. [1] The coefficient of x^ν is different from zero, and therefore x^ν can also be expressed by $B_\nu^{(x)}$ and by lower powers of x. Thus every whole function of degree n in x can also be linearly expressed by $B_0^{(x)}, B_1^{(x)}, \dots, B_n^{(x)}$ in the form

$$f(x) = M_0B_0^{(x)} + M_1B_1^{(x)} + \cdots + M_nB_n^{(x)}, \tag{4}$$

where $M_0, M_1, \dots, M_n$ are constants; and the function $f(x)$ is determined when these constants are determined.

Now, by our supposition, $f(0), f(1), f(2), \dots, f(n)$ are given. Since, by (3), $B_\nu^{(x)}$ always vanishes when x has one of the values $0, 1, 2, \dots, \nu-1$, the following linear equations for the unknowns M are obtained from (4):

$$\begin{aligned} f(0) &= M_0B_0^{(0)}, \\ f(1) &= M_0B_0^{(1)} + M_1B_1^{(1)}, \\ f(2) &= M_0B_0^{(2)} + M_1B_1^{(2)} + M_2B_2^{(2)}, \\ &\cdots \\ f(n) &= M_0B_0^{(n)} + M_1B_1^{(n)} + \cdots + M_nB_n^{(n)}. \end{aligned} \tag{5}$$

These equations are now to be solved with respect to $M_0, M_1, \dots, M_n$, which is done very easily with the help of the final equations (8) of the preceding paragraph. The first equation (5) gives directly

$$M_0 = f(0). \tag{6}$$

[1] The meaning of these generalized binomial coefficients for the expansion of powers of a binomial does not belong here, but to analysis.

If the first equation (5) is multiplied by $B_0^{(1)}$, the second by $-B_1^{(1)}$, and they are added, one obtains, by the formula system mentioned (applied to $n = 1$),

$$-M_1 = B_0^{(1)} f(0) - B_1^{(1)} f(1),$$

and so in general, if the first ν equations (5) are multiplied in order by

$$B_0^{(\nu)}, \quad -B_1^{(\nu)}, \quad +B_2^{(\nu)}, \quad \dots, \quad \pm B_\nu^{(\nu)}$$

and then added,

$$\pm M_\nu = B_0^{(\nu)} f(0) - B_1^{(\nu)} f(1) + B_2^{(\nu)} f(2) - \cdots \pm B_\nu^{(\nu)} f(\nu), \tag{7}$$

whereby, according to (4), the function $f(x)$ is determined and the problem is solved. It is clear that, so long as we make no special supposition about the values $f(0), f(1), \dots, f(n)$, every arbitrary whole rational function of x can be represented in this form.

§ 10. Solution of the interpolation problem by differences.

The definition of the whole rational function

$$B_\nu^{(x)} = \frac{x(x-1)\cdots(x-\nu+1)}{1 \cdot 2 \cdot 3 \cdots \nu} \tag{1}$$

gives the relation, already proved earlier for the special case of a positive integral x,

$$B_\nu^{(x+1)} = B_\nu^{(x)} + B_{\nu-1}^{(x)}, \qquad B_0^{(x+1)} = B_0^{(x)} = 1. \tag{2}$$

From this we obtain for the coefficients $M_0, M_1, \dots, M_n$ a mode of determination which is much more convenient for practical calculation than the application of the formulae of the preceding paragraph.

Namely, by the formulae (4) and (6) of § 9,

$$f(x) = f(0) + M_1 B_1^{(x)} + M_2 B_2^{(x)} + \cdots + M_n B_n^{(x)}, \tag{3}$$

and if in this expression we replace x by $x + 1$ and form the difference

$$\Delta_x = f(x+1) - f(x), \tag{4}$$

then, with regard to (2),

$$\Delta_x = M_1 + M_2 B_1^{(x)} + M_3 B_2^{(x)} + \cdots + M_n B_{n-1}^{(x)}, \tag{5}$$

from which follows (since (5) is an equation of the same kind as (3), except that $n - 1$ has taken the place of n)

$$M_1 = \Delta_0 = f(1) - f(0).$$

Put

$$\Delta'_x = \Delta_{x+1} - \Delta_x, \qquad \Delta''_x = \Delta'_{x+1} - \Delta'_x, \quad \dots, \tag{6}$$

then, accordingly,

$$M_0 = f(0), \quad M_1 = \Delta_0, \quad M_2 = \Delta'_0, \quad \dots, \quad M_n = \Delta_0^{(n-1)},$$

and formula (3) gives

$$f(x) = f(0) + \Delta_0 B_1^{(x)} + \Delta'_0 B_2^{(x)} + \Delta''_0 B_3^{(x)} + \cdots + \Delta_0^{(n-1)} B_n^{(x)}. \tag{7}$$

Likewise the functions $\Delta_x, \Delta'_x, \Delta''_x, \ldots$ can be expressed:

$$\begin{aligned} \Delta_x &= \Delta_0 + \Delta'_0 B_1^{(x)} + \Delta''_0 B_2^{(x)} + \cdots + \Delta_0^{(n-1)} B_{n-1}^{(x)}, \\ \Delta'_x &= \Delta'_0 + \Delta''_0 B_1^{(x)} + \cdots + \Delta_0^{(n-1)} B_{n-2}^{(x)}, \end{aligned} \tag{8}$$

and so on.

The $\Delta_x, \Delta'_x, \Delta''_x, \ldots, \Delta_x^{(n-1)}$ are whole rational functions of degrees $n-1, n-2, \ldots, 0$; the last of them is therefore constant. In order to represent them all, one needs only the values

$$f(0), \quad \Delta_0, \quad \Delta'_0, \quad \Delta''_0, \quad \ldots, \quad \Delta_0^{(n-1)},$$

which one computes most easily by arranging a table. For the case $n = 3$, for example, the table has the form

$$\begin{array}{llll} f(0) & & & \\ f(1) & \Delta_0 & & \\ f(2) & \Delta_1 & \Delta'_0 & \\ f(3) & \Delta_2 & \Delta'_1 & \Delta''_0 \end{array}$$

where the differences are to be formed diagonally.

§ 11. Arithmetic series of higher order.

The preceding considerations can also be stated in another form, which we shall need later. If

$$u_0, u_1, u_2, \ldots \tag{1}$$

is a sequence of quantities, we call

$$\Delta_0 = u_1 - u_0, \Delta_1 = u_2 - u_1, \Delta_2 = u_3 - u_2, \ldots \tag{2}$$

the sequence of its differences; and

$$\Delta'_0 = \Delta_1 - \Delta_0, \Delta'_1 = \Delta_2 - \Delta_1, \ldots \tag{3}$$

the sequence of its second differences, and so on.

It is clear that the sequence (1) is completely determined when its first term and the sequence of its first differences are given; for

$$u_1 = u_0 + \Delta_0, \quad u_2 = u_0 + \Delta_0 + \Delta_1, \quad \ldots,$$

$$u_m = u_0 + \Delta_0 + \Delta_1 + \cdots + \Delta_{m-1}.$$

Likewise the sequence (1) is completely determined when the first two terms and the sequence of its second differences are given, and so forth. The sequence (1) is called an arithmetic series of nth order when the sequence of its nth differences is constant, and hence the sequence of the $(n+1)$st differences consists entirely of zeros.

One obtains an arithmetic series of nth order by substituting the numbers $0, 1, 2, 3, \ldots$ for x in a whole function $f(x)$ of degree n. For if one puts

$$\Delta_x = f(x+1) - f(x), \quad \Delta_x^{(n-1)} = \Delta_{x+1}^{(n-2)} - \Delta_x^{(n-2)},$$

then Δ_x is of degree $n-1$, Δ'_x of degree $n-2$, and so on with respect to x, and therefore $\Delta_x^{(n-1)}$ is constant.

If now

$$u_0, u_1, u_2, \ldots$$

is an arithmetic series of nth order, then the whole series is completely determined when the n values $u_0, u_1, u_2, \ldots, u_{n-1}$ and, besides these, the constant nth difference are given. But this latter is determined if also the $(n+1)$st term u_n is given. We can therefore state the theorem:

An arithmetic series of nth order is completely determined when its first $n+1$ terms are given.

Since a function $f(x)$ of degree n is likewise completely determined by the arbitrarily given values

$$f(0), f(1), f(2), \ldots, f(n),$$

it follows that from the whole rational functions of degree n all arithmetic series of nth order are produced when in them the series of natural numbers is substituted for x.

The expression for the general term is then given by formula (7) of the preceding paragraph.

The sums of the first $m+1$ terms of an arithmetic series of nth order,

$$s_m = u_0 + u_1 + \cdots + u_m,$$

form an arithmetic series of $(n+1)$st order, since their first differences

$$s_{m+1} - s_m = u_{m+1}$$

form an arithmetic series of nth order.

Thus, with the help of formula (7) of § 10, the sum s_m can be determined in general when $s_0, s_1, \ldots, s_{n+1}$ are assumed known.

To find the generating function $F(x)$ of s_m, when $f(x)$ is the generating function of u_m, put

$$F(0) = f(0), \quad F(1) = f(0) + f(1), \quad F(2) = f(0) + f(1) + f(2), \ldots$$

and in formula (7), § 10,

$$F(x) = F(0) + D_0 B_1^{(x)} + D_0' B_2^{(x)} + \cdots$$

one has to put

$$\begin{aligned} D_0 &= F(1) - F(0) = f(1), & D_0' &= f(2) - f(1) = \Delta_1, \\ D_1 &= F(2) - F(1) = f(2), & D_1' &= f(3) - f(2) = \Delta_2, \\ D_2 &= F(3) - F(2) = f(3), & D_0'' &= \Delta_2 - \Delta_1 = \Delta_1', \end{aligned}$$

and so on. Thus one obtains

$$F(x) = f(0) + f(1) B_1^{(x)} + \Delta_1 B_2^{(x)} + \Delta_1' B_3^{(x)} + \cdots. \tag{4}$$

For example, take $f(x) = x^2$; then $F(m)$ gives the sum of the first m square numbers. Here

$$\Delta_x = 2x + 1, \qquad \Delta_x' = 2,$$

and therefore

$$F(x) = x + 3\frac{x(x-1)}{1 \cdot 2} + 2\frac{x(x-1)(x-2)}{1 \cdot 2 \cdot 3} = \frac{x(x+1)(2x+1)}{6}.$$

For $f(x) = x^3$ the same calculation gives

$$F(x) = \left(\frac{x(x+1)}{2}\right)^2.$$

Thus the sum of the first m cubes is equal to the square of the mth triangular number.

§ 12. The polynomial theorem.

In § 8 the following form of the binomial theorem was derived:

$$(x+y)^n = \Pi(n) \sum^{\alpha,\beta} \frac{x^\alpha y^\beta}{\Pi(\alpha)\Pi(\beta)}, \tag{1}$$

where the sum extends over all combinations of two numbers α, β, neither of which is negative, and which satisfy the condition

$$\alpha + \beta = n. \tag{2}$$

This form permits, first by induction, a generalization to the nth power of a polynomial:

$$(x+y+z+\cdots)^n = \Pi(n) \sum^{\alpha,\beta,\gamma,\ldots} \frac{x^\alpha y^\beta z^\gamma \cdots}{\Pi(\alpha)\Pi(\beta)\Pi(\gamma)\cdots}, \tag{3}$$

with the determination that $\alpha, \beta, \gamma, \ldots$ run through all positive or vanishing integral values which satisfy

$$\alpha + \beta + \gamma + \cdots = n. \tag{4}$$

But to prove the correctness of this formula in general, we assume that it is proved when the polynomial contains one term fewer, as is indeed the case when the polynomial contains only two terms.

We then put

$$u = y + z + \cdots \tag{5}$$

and apply formula (1) to $(x+u)$, from which there follows

$$(x+y+z+\cdots)^n = \Pi(n) \sum^{\alpha,\nu} \frac{x^\alpha u^\nu}{\Pi(\alpha)\Pi(\nu)}, \tag{6}$$

with the restriction

$$\alpha + \nu = n. \tag{7}$$

By the assumption, however, it has already been proved that

$$u^\nu = \Pi(\nu) \sum^{\beta,\gamma,\ldots} \frac{y^\beta z^\gamma \cdots}{\Pi(\beta)\Pi(\gamma)\cdots}, \tag{8}$$

where

$$\beta + \gamma + \cdots = \nu. \tag{9}$$

When this is inserted in (6), formula (3) is obtained immediately, and (7) passes over into (4).

The coefficients

$$P^{(n)}_{\alpha,\beta,\gamma,\ldots} = \frac{\Pi(n)}{\Pi(\alpha)\Pi(\beta)\Pi(\gamma)\cdots}, \tag{10}$$

which, according to their meaning, are whole numbers, are called the polynomial coefficients.

For example, for the third power of the trinomial one obtains

$$\begin{aligned}(x+y+z)^3 = x^3 + y^3 + z^3 + 3x^2y + 3xy^2 + 3x^2z + 3xz^2 \\ + 3y^2z + 3yz^2 + 6xyz.\end{aligned} \tag{11}$$